

TUNG DUONG

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🐙 github.com/tungduong2

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science; Specialization in Artificial Intelligence; GPA: 4.0/4.0

May 2026

Atlanta, GA

University of Massachusetts Boston

Bachelor of Science in Computer Science; Summa Cum Laude - Latin Honors; GPA: 3.92/4.0

May 2022

Boston, MA

Bunker Hill Community College

Associate of Science in Computer Support Specialist; High Honors; GPA: 3.85/4.0

May 2020

Boston, MA

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, PHP, C#, C++.

Data, Cloud & Backend: AWS, Azure CLI, Docker, MySQL, REST APIs, JSON, Node, React, Symfony.

Technologies & Tools: Git, GitHub, VM, VSCode, IntelliJ, PyCharm, Linux CLI, Data Modeling.

Eligibility status: I have authorization to work permanently in the United States.

EXPERIENCE

Applications Software Developer

University of Miami

Nov 2022 – Present

Coral Gables, FL (Remote)

- Architect backend API integrations and data ingestion pipelines for enterprise platforms, utilizing Python and Azure AI Search protocols for large-scale metadata harvesting.
- Modernize legacy infrastructure by upgrading core systems to PHP 8.3 and MySQL 8.4 LTS, and overhauling Docker architectures to separate build/runtime libraries for optimized container efficiency.
- Lead database migrations and structural cleanups, refactoring complex ManyToMany relationships into OneToMany arrays to enforce data integrity and reduce query latency.
- Implement enterprise-level security by integrating Microsoft Entra ID (Azure SSO) across organizational Intranets and systematically resolving high-severity Dependabot vulnerabilities.

Office365 and System Administrator Apprenticeship

University of Massachusetts Boston

Aug 2020 – Jun 2022

Boston, MA

- Managed active directory lifecycles and large-scale data reconciliation as part of the Professional Apprenticeship and Career Experience (PACE) program.
- Automated enterprise data processing operations using scripting and FileMaker Pro, significantly reducing manual ticket resolution time for communication pipelines.

Computer Technician

LCT Computer

Oct 2016 – Oct 2018

Rockland, MA

- Troubleshoot hardware, OS, and software issues for desktop and enterprise systems, specializing in custom builds and hardware optimization.

PROJECTS [🔗 VIEW FULL PORTFOLIO HERE](#)

U.S. Housing Market Analytics Platform | *Python, D3.js, Data Modeling, Clustering*

2025-2026

- Engineered a multi-scale, interactive web platform visualizing the impact of corporate investor activity on the U.S. housing market by integrating Zillow time-series data and Redfin investor metrics.
- Implemented distance-based K-Means clustering to categorize city-level investor trends and utilized Holt-Winters exponential smoothing models to forecast future housing prices.

Patient-First Interoperable Portal | *Python (FastAPI), React, JSON, REST APIs*

2025-2026

- Architected a FHIR-compliant full-stack patient-provider portal that prioritizes patient agency and control over highly sensitive health data and medical charting.
- Engineered secure backend APIs allowing patients to selectively authorize and share specific subsets of data (medications, conditions) with providers via generated authentication codes.

ARC-AGI Grid Transformation Agent | *Python, Machine Learning, Search Algorithms*

2025-2026

- Developed an AI agent that successfully solved 91% of complex grid transformation problems by discovering advanced transformation sequences to convert input meshes into target outputs.
- Integrated a custom transformation library with a beam search algorithm, a heuristic scoring system, and training set parameter inference to optimize logic paths.

Subjects Plus [🔗](#) | *Symfony, PHP, MySQL, API, Docker, Azure*

2023-2026

- Lead critical backend refactoring for SubjectsPlus, an open-source enterprise database tool, successfully upgrading to Symfony 6.4 and converting 130+ core files to modern PHP attributes.
- Maintain distribution codebases and manage system security audits, systematically resolving over 45 critical dependency vulnerabilities to ensure application stability.

AWARDS AND HONORS

Code4Lib 2023 Diversity Scholarship Award

Code4Lib Conference at Princeton University

2023

Princeton, NJ

Paul M. English '87 Computer Science Endowed Scholarship (\$10,000)

University of Massachusetts Boston

2021-2022

Boston, MA